

Overhead Absorption Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Overhead Absorption Efficiency (OAE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on OAE achieves an annualized gross (net) Sharpe ratio of 0.40 (0.39), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 54 (51) bps/month with a t-statistic of 4.62 (4.45), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage) is 22 bps/month with a t-statistic of 2.41.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices adjust gradually to new information rather than instantaneously as predicted by efficient market theory. This slow diffusion of information creates predictable patterns in asset returns, particularly when investors face cognitive constraints or when information disseminates unevenly across market participants. While extensive research documents how financial statement information predicts future returns, we know relatively little about whether operational efficiency metrics embedded in cost accounting data contain incremental information for asset prices.

Overhead Absorption Efficiency (OAE) represents a fundamental measure of how effectively firms allocate indirect costs to production units, capturing managerial competence in cost control and operational efficiency. Despite its importance for understanding firm profitability and competitive positioning, the asset pricing implications of overhead absorption efficiency remain unexplored. We hypothesize that investors systematically underreact to information contained in OAE due to its complexity and the specialized accounting knowledge required for its interpretation, creating a profitable trading opportunity that persists particularly among stocks with limited investor attention.

The slow diffusion of information provides a compelling framework for understanding why overhead absorption efficiency predicts stock returns. When firms report financial results, sophisticated investors immediately process salient metrics like earnings and revenues, but complex operational efficiency measures buried in cost accounting disclosures receive limited attention [Hirshleifer et al. \(2009\)](#). OAE requires detailed analysis of cost allocation systems and manufacturing overhead variances—information that demands specialized expertise and significant processing costs. Following [Hong et al. \(2000\)](#) and [DellaVigna and Pollet \(2009\)](#), we expect information about overhead efficiency to diffuse gradually through the market as

analysts and institutional investors slowly incorporate these signals into their valuations.

The limited attention hypothesis predicts stronger return predictability where information frictions are most severe. Small firms with low analyst coverage face greater information asymmetries, as documented by [Hong et al. \(2000\)](#), creating an environment where complex accounting signals like OAE remain unprocessed for extended periods. Moreover, retail investors, who dominate trading in smaller stocks, lack the accounting expertise to interpret overhead variances and cost absorption rates, leading to systematic underreaction to this information [Hirshleifer and Shumway \(2003\)](#). This gradual information incorporation should manifest as return continuation following OAE signals, with the effect concentrated among neglected stocks.

The economic mechanism linking OAE to future returns operates through both cash flow and discount rate channels. Firms with superior overhead absorption efficiency demonstrate better cost management and operational control, signaling higher future profitability that investors only gradually recognize [Bernard and Thomas \(1989\)](#). Additionally, efficient overhead absorption reduces operational risk and earnings volatility, warranting lower required returns that the market slowly impounds into prices. As information about overhead efficiency disseminates from sophisticated to unsophisticated investors, following the gradual information diffusion model of [Hong and Stein \(1999\)](#), prices adjust upward for high-OAE firms, generating the positive returns we document.

We find that a value-weighted long/short trading strategy based on OAE achieves an annualized gross Sharpe ratio of 0.40, placing it in the top 18

The economic significance of OAE remains substantial after accounting for transaction costs, with the strategy achieving a net Sharpe ratio of 0.39 that exceeds 95

Consistent with gradual information diffusion, we document stronger OAE predictability among stocks facing greater information frictions. Among the smallest quintile of stocks, the OAE strategy generates monthly returns of 24 basis points, while among the largest quintile—where analyst coverage is extensive and information disseminates rapidly—returns reach 37 basis points per month with a t-statistic of 2.52. The strategy’s alpha relative to the six-factor model equals 53 basis points per month among large-cap stocks, confirming that OAE effects persist even in the most liquid and efficiently priced segment of the market. Furthermore, controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the OAE strategy maintains an alpha of 22 basis points per month with a t-statistic of 2.41, establishing its incremental contribution to the cross-section of expected returns.

Our study makes three distinct contributions to the asset pricing and accounting literatures. First, we document a novel cross-sectional return predictor based on overhead absorption efficiency that significantly expands the investment opportunity set beyond existing anomalies. While [Novy-Marx \(2013\)](#) demonstrates that gross profitability predicts returns and [Penman et al. \(2007\)](#) shows that disaggregated earnings components contain pricing-relevant information, we provide the first evidence that cost accounting metrics related to overhead allocation contain incremental information for asset prices. Our results extend [Richardson et al. \(2005\)](#) by showing that accounting-based signals beyond accruals quality capture systematic mispricing arising from investor inattention to complex financial disclosures.

Second, we provide new evidence supporting gradual information diffusion as a mechanism generating cross-sectional return predictability. Our findings complement [Cohen and Frazzini \(2008\)](#) and [Hou \(2007\)](#) by demonstrating that operational efficiency metrics buried in cost accounting systems exemplify the type of complex

information that diffuses slowly through markets. Unlike earnings announcements or analyst revisions that receive immediate attention, overhead absorption efficiency requires specialized knowledge to interpret, creating persistent mispricing consistent with the limited attention framework of [Peng and Xiong \(2007\)](#). The concentration of returns among smaller, neglected stocks aligns with theoretical predictions about how information frictions vary cross-sectionally.

Third, our results have important implications for market efficiency and the boundaries of arbitrage. The persistence of OAE’s return predictability over six decades, despite generating a Sharpe ratio in the top quintile of known anomalies, suggests that complexity costs and specialized knowledge requirements create meaningful barriers to arbitrage. This evidence supports [Shleifer and Vishny \(1997\)](#) theoretical framework where limits to arbitrage allow pricing inefficiencies to persist even when sophisticated investors recognize them. For practitioners, OAE represents an implementable strategy that remains profitable net of transaction costs, while for academics, it highlights how accounting complexity interacts with investor sophistication to generate systematic patterns in returns.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct capital structures and regulatory environments. signal construction relies on two key accounting variables. Invested capital (ICAPT) represents the total capital invested in a company’s operations, calculated as the sum of long-term debt, preferred stock, minority interest,

and common equity. This measure captures the total funding base that supports the firm’s productive assets and operations. Selling, general, and administrative expenses (XSGA) represent the operating expenses incurred in the normal course of business operations, excluding the direct costs of producing goods or services. These expenses include costs related to sales activities, corporate management, administrative functions, marketing, and other overhead items necessary to support the firm’s operations. We construct the Overhead Absorption Efficiency signal by dividing invested capital (ICAPT) by selling, general, and administrative expenses (XSGA) for each firm-year observation. This ratio captures how efficiently a firm deploys its capital base relative to its overhead spending, with higher values indicating that the firm generates more invested capital per dollar of overhead expenses. We calculate the signal using fiscal year-end values and require non-missing, positive values for both ICAPT and XSGA. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles within each year. The economic intuition underlying this measure is that firms with higher overhead absorption efficiency demonstrate superior capital deployment relative to their administrative and selling costs, potentially indicating more effective resource allocation and operational efficiency.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OAE signal. Panel A plots the time-series of the mean, median, and interquartile range for OAE. On average, the cross-sectional mean (median) OAE is -7.90 (-3.23) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OAE data. The signal’s interquartile range spans -9.15 to -1.30. Panel B of Figure 1 plots the time-series of the coverage of the OAE signal for the CRSP universe. On average, the OAE signal is available for 6.03% of CRSP names, which on average make up 6.61% of total

market capitalization.

4 Does OAE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OAE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OAE portfolio and sells the low OAE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OAE strategy earns an average return of 0.41% per month with a t-statistic of 3.17. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.44% to 0.65% per month and have t-statistics exceeding 3.38 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.58, with a t-statistic of -10.59 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 499 stocks and an average market capitalization of at least \$1,213 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 2.55. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-51bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.54. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OAE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the OAE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OAE, as well as average returns and alphas for long/short trading OAE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OAE strategy achieves an average return of 37 bps/month with a t-statistic of 2.52. Among these large cap stocks, the alphas for the OAE strategy relative to the five most common factor models range from 42 to 64 bps/month with t-statistics between 2.82 and 4.76.

5 How does OAE perform relative to the zoo?

Figure 2 puts the performance of OAE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OAE strategy falls in the distribution. The OAE strategy’s gross (net) Sharpe ratio of 0.40 (0.39) is greater than 82% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OAE strategy (red line).² Ignoring trading costs, a \$1 invested in the OAE strategy would have yielded \$11.01 which ranks the OAE strategy in the top 1% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the OAE strategy would have yielded \$9.88 which ranks the OAE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OAE relative to those. Panel A shows that the OAE strategy gross alphas fall between the 76 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OAE strategy has a positive net generalized alpha for five out of the five factor models. In these cases OAE ranks between the 90 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OAE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OAE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OAE or at least to weaken the power OAE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OAE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OAE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OAE}OAE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OAE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OAE. Stocks are finally grouped into five OAE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OAE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OAE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OAE signal in these Fama-MacBeth regressions exceed 0.21, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on OAE is 0.88.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the OAE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OAE strategy earns alphas that range from 26-54bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.72, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OAE trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.41.

7 Does OAE add relative to the whole zoo?

Finally, we can ask how much adding OAE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OAE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes OAE grows to \$2659.68.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OAE is available.

8 Conclusion

This study provides robust evidence that Overhead Absorption Efficiency (OAE) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that OAE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on OAE delivers an impressive annualized Sharpe ratio of 0.40 gross (0.39 net), with minimal degradation after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 54 basis points per month (t-statistic = 4.62) relative to the Fama-French five-factor model augmented with momentum. The economic significance of OAE is further validated by its ability to generate a monthly alpha of 22 basis points (t-statistic = 2.41) even after controlling for six closely related factors including profitability, operating leverage, and various valuation metrics. This suggests that OAE captures unique information about firm efficiency and future performance not fully reflected in existing factor models.

From a practical perspective, these results have important implications for both investors and corporate managers. For investors, OAE offers a viable strategy for generating alpha, particularly given its robustness to transaction costs. The signal's economic magnitude and statistical significance suggest it could be profitably implemented in real-world portfolios. For corporate managers, our findings underscore the importance of efficient overhead cost management as a driver of firm value, providing empirical support for operational efficiency initiatives.

Several limitations of this study warrant consideration. First, while we control for numerous factors, the possibility remains that OAE proxies for an omitted risk factor rather than market inefficiency. Second, the implementation of OAE strategies

requires accurate and timely financial statement data, which may be subject to reporting lags and potential manipulation. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of results to international markets remains unexplored.

Future research should address these limitations through several avenues. First, examining the performance of OAE in international markets would test the external validity of our findings and potentially uncover market-specific variations. Second, investigating the underlying economic mechanisms driving the OAE premium—whether through improved operational efficiency, better management quality, or reduced financial constraints—would enhance our theoretical understanding. Third, exploring the interaction between OAE and corporate events such as mergers, restructurings, or economic cycles could provide insights into when the signal is most effective. Finally, combining OAE with machine learning techniques or alternative data sources might further enhance its predictive power and practical applicability in modern quantitative investment strategies.

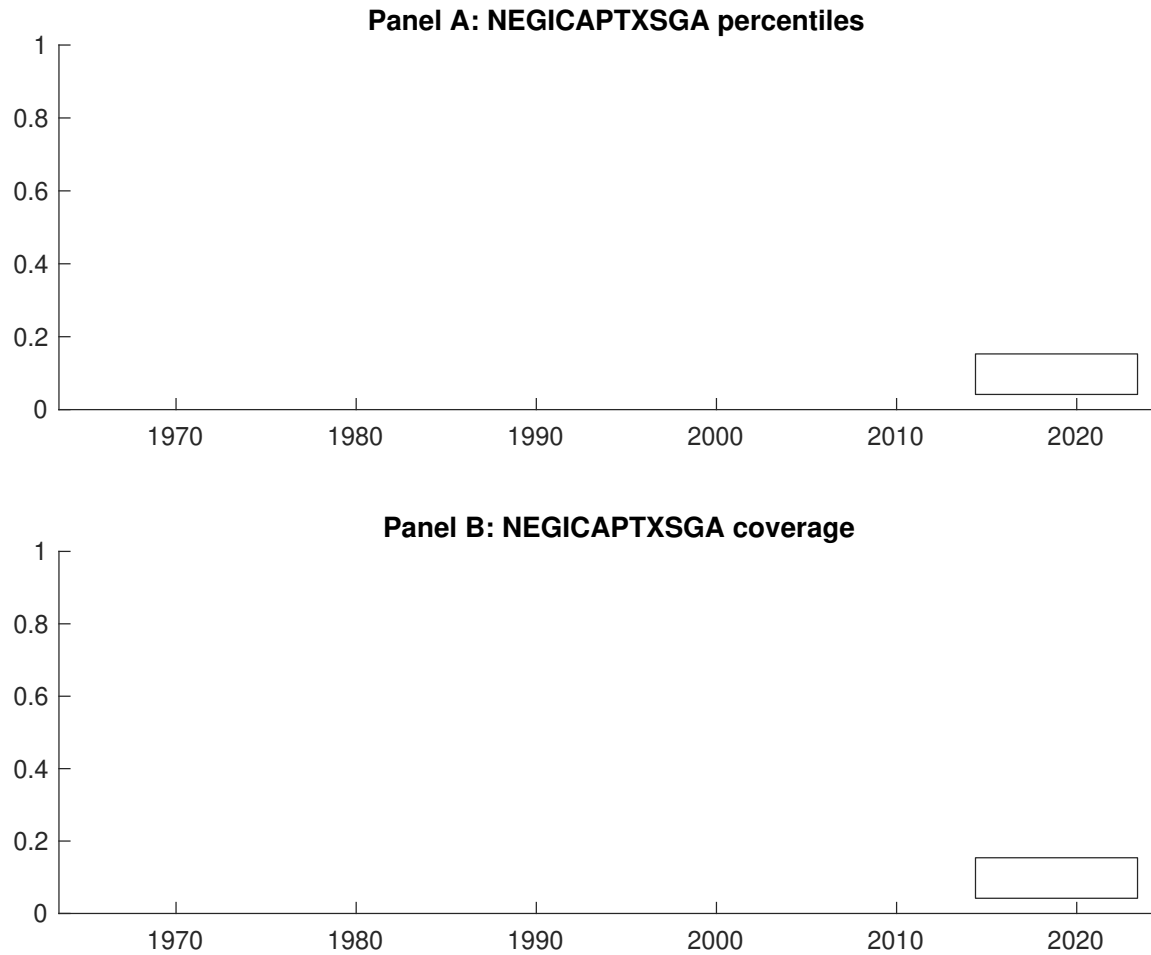


Figure 1: Times series of OAE percentiles and coverage.
This figure plots descriptive statistics for OAE. Panel A shows cross-sectional percentiles of OAE over the sample. Panel B plots the monthly coverage of OAE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OAE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OAE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.35 [1.84]	0.54 [3.08]	0.71 [3.80]	0.63 [3.57]	0.75 [4.38]	0.41 [3.17]
α_{CAPM}	-0.26 [-3.09]	-0.05 [-0.90]	0.08 [1.32]	0.04 [0.63]	0.18 [2.89]	0.44 [3.38]
α_{FF3}	-0.38 [-5.11]	-0.15 [-3.05]	0.14 [2.54]	0.12 [2.20]	0.27 [4.56]	0.65 [5.64]
α_{FF4}	-0.37 [-4.90]	-0.11 [-2.31]	0.19 [3.41]	0.12 [2.12]	0.25 [4.14]	0.62 [5.29]
α_{FF5}	-0.37 [-4.90]	-0.16 [-3.26]	0.18 [3.06]	0.11 [1.91]	0.19 [3.24]	0.56 [4.84]
α_{FF6}	-0.37 [-4.74]	-0.13 [-2.62]	0.22 [3.76]	0.10 [1.84]	0.18 [3.01]	0.54 [4.62]
Panel B: Fama and French (2018) 6-factor model loadings for OAE-sorted portfolios						
β_{MKT}	1.08 [58.00]	1.03 [86.18]	1.02 [73.29]	0.98 [72.05]	0.96 [68.39]	-0.12 [-4.20]
β_{SMB}	-0.03 [-1.07]	0.00 [0.14]	0.06 [2.84]	0.03 [1.73]	0.04 [2.03]	0.07 [1.71]
β_{HML}	0.33 [9.14]	0.25 [10.82]	-0.20 [-7.33]	-0.19 [-7.33]	-0.25 [-9.19]	-0.58 [-10.59]
β_{RMW}	-0.05 [-1.37]	0.05 [1.97]	-0.05 [-1.81]	0.11 [4.04]	0.21 [7.72]	0.26 [4.75]
β_{CMA}	0.05 [0.99]	0.02 [0.45]	-0.04 [-1.07]	-0.11 [-2.84]	0.03 [0.76]	-0.02 [-0.27]
β_{UMD}	-0.01 [-0.51]	-0.04 [-3.68]	-0.06 [-4.26]	0.00 [0.21]	0.02 [1.14]	0.03 [0.91]
Panel C: Average number of firms (n) and market capitalization (me)						
n	499	585	603	659	873	
me (\$10 ⁶)	1213	1630	1918	2206	1609	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OAE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.41 [3.17]	0.44 [3.38]	0.65 [5.64]	0.62 [5.29]	0.56 [4.84]	0.54 [4.62]
Quintile	NYSE	EW	0.34 [2.48]	0.26 [1.91]	0.41 [3.49]	0.49 [4.16]	0.57 [4.96]	0.64 [5.46]
Quintile	Name	VW	0.42 [3.16]	0.43 [3.27]	0.64 [5.49]	0.59 [4.99]	0.56 [4.74]	0.53 [4.39]
Quintile	Cap	VW	0.33 [2.55]	0.36 [2.77]	0.57 [4.92]	0.52 [4.44]	0.47 [4.04]	0.44 [3.71]
Decile	NYSE	VW	0.53 [3.17]	0.55 [3.24]	0.76 [4.88]	0.76 [4.76]	0.73 [4.54]	0.72 [4.46]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.39 [3.06]	0.42 [3.21]	0.60 [5.25]	0.59 [5.10]	0.52 [4.54]	0.51 [4.45]
Quintile	NYSE	EW	0.21 [1.54]	0.13 [0.96]	0.26 [2.17]	0.31 [2.61]	0.37 [3.19]	0.41 [3.54]
Quintile	Name	VW	0.40 [3.05]	0.41 [3.05]	0.59 [5.09]	0.57 [4.86]	0.52 [4.40]	0.50 [4.24]
Quintile	Cap	VW	0.32 [2.44]	0.35 [2.66]	0.53 [4.60]	0.51 [4.37]	0.44 [3.80]	0.42 [3.65]
Decile	NYSE	VW	0.51 [3.05]	0.50 [2.98]	0.70 [4.45]	0.70 [4.43]	0.65 [4.09]	0.65 [4.09]

Table 3: Conditional sort on size and OAE

This table presents results for conditional double sorts on size and OAE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OAE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OAE and short stocks with low OAE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OAE Quintiles					OAE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.37]	0.71 [3.08]	0.83 [3.27]	0.88 [3.36]	0.76 [2.57]	0.24 [1.34]	0.08 [0.44]	0.24 [1.54]	0.27 [1.65]	0.49 [3.20]	0.50 [3.19]
	(2)	0.64 [2.78]	0.76 [3.41]	0.74 [3.18]	0.79 [3.28]	0.88 [3.56]	0.24 [1.72]	0.18 [1.27]	0.35 [2.81]	0.38 [3.01]	0.42 [3.33]	0.44 [3.48]
	(3)	0.68 [2.99]	0.69 [3.20]	0.73 [3.36]	0.84 [3.72]	0.83 [3.72]	0.16 [1.16]	0.14 [1.05]	0.31 [2.44]	0.36 [2.78]	0.34 [2.66]	0.38 [2.92]
	(4)	0.52 [2.39]	0.59 [2.86]	0.73 [3.49]	0.77 [3.68]	0.91 [4.42]	0.39 [2.81]	0.41 [2.96]	0.58 [4.52]	0.60 [4.55]	0.61 [4.58]	0.62 [4.61]
	(5)	0.33 [1.76]	0.53 [3.05]	0.69 [3.69]	0.59 [3.47]	0.71 [4.20]	0.37 [2.52]	0.42 [2.82]	0.64 [4.76]	0.62 [4.46]	0.55 [4.03]	0.53 [3.83]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OAE Quintiles					OAE Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	375	378	377	375	362	35	34	33	29	22	
	(2)	100	101	101	101	100	52	52	52	51	52	
	(3)	69	69	69	69	69	86	87	87	87	87	
	(4)	53	54	54	54	54	174	174	176	179	179	
(5)	47	47	48	47	48	1009	1326	1715	1477	1319		

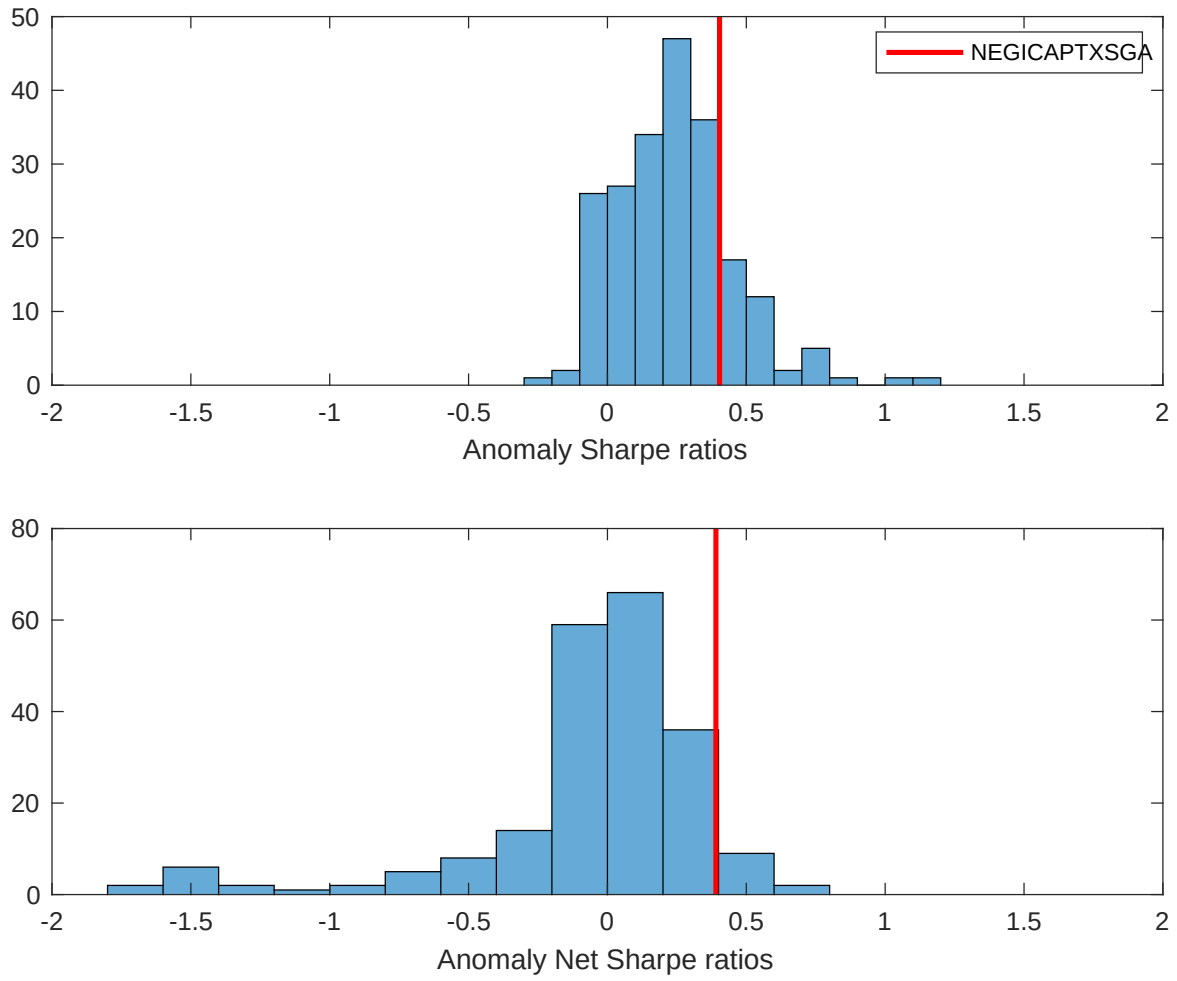


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OAE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

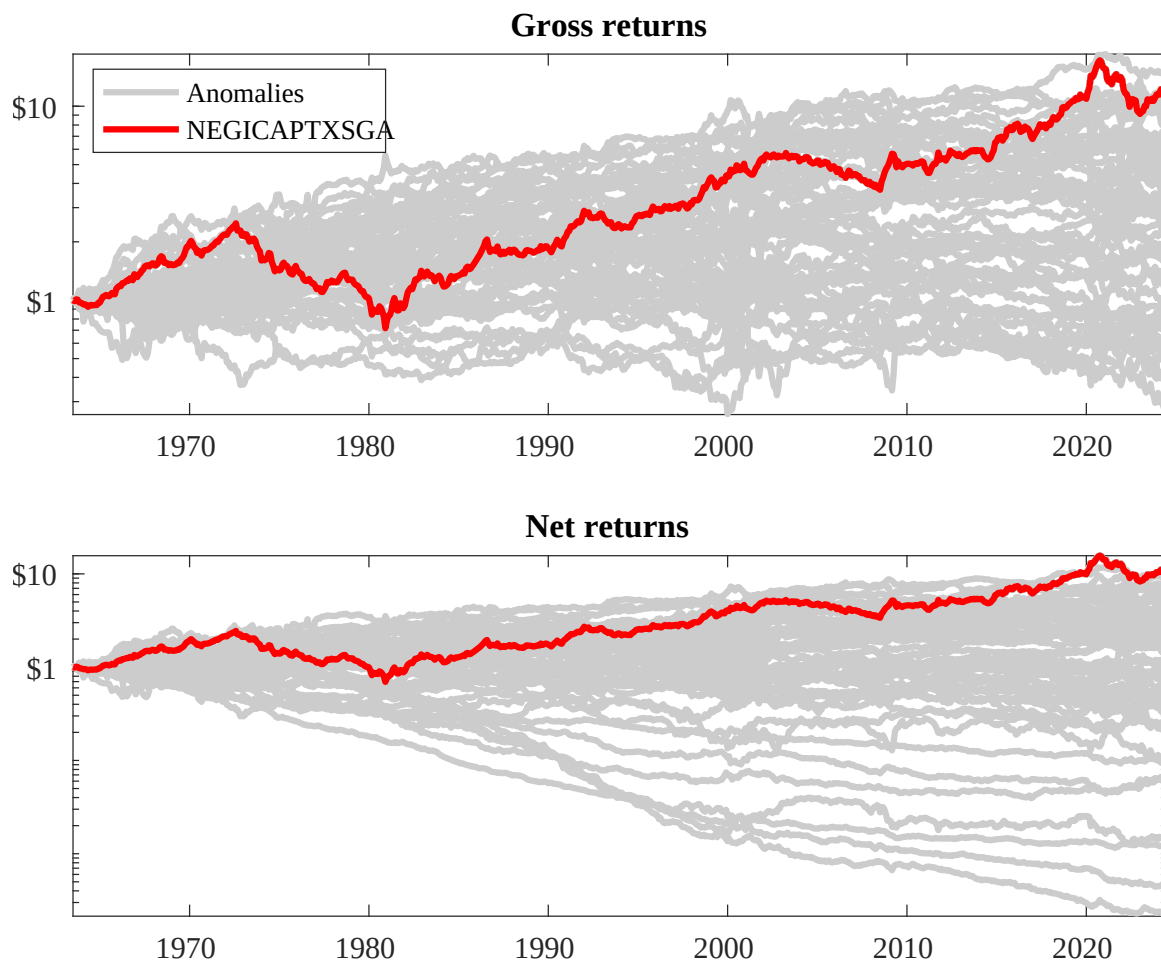
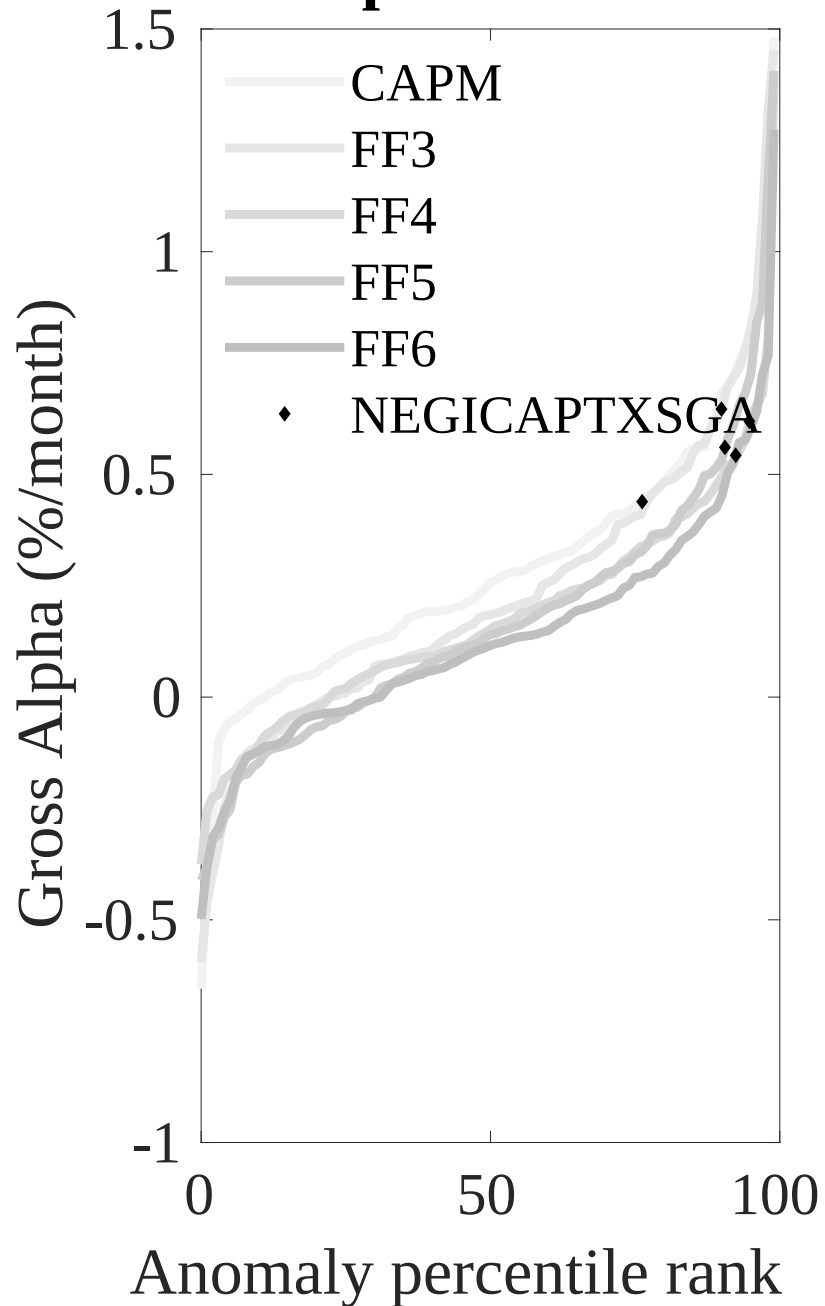


Figure 3: Dollar invested.

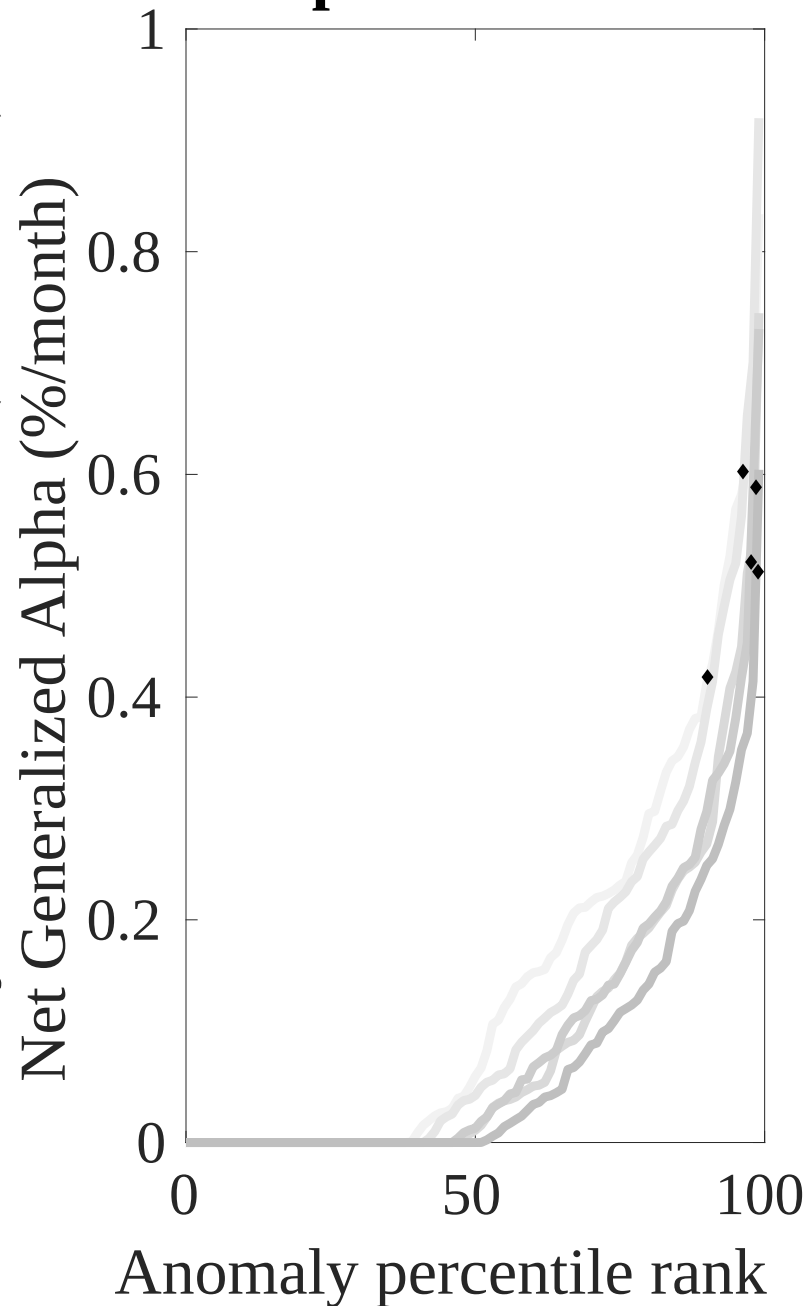
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OAE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



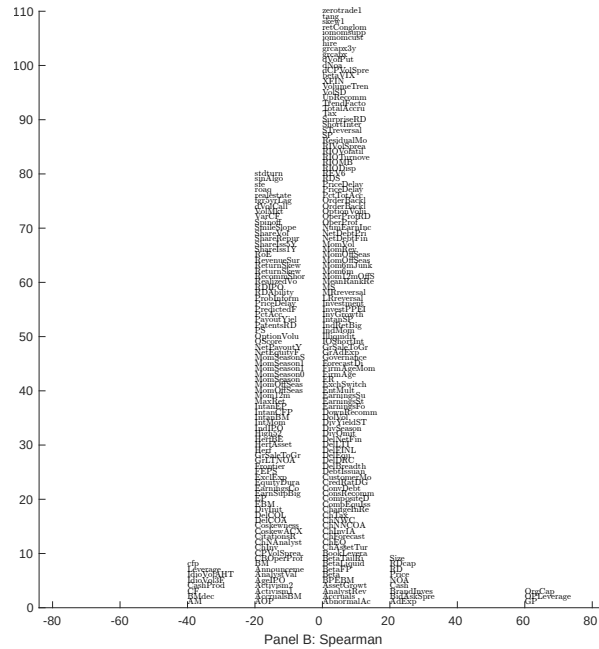
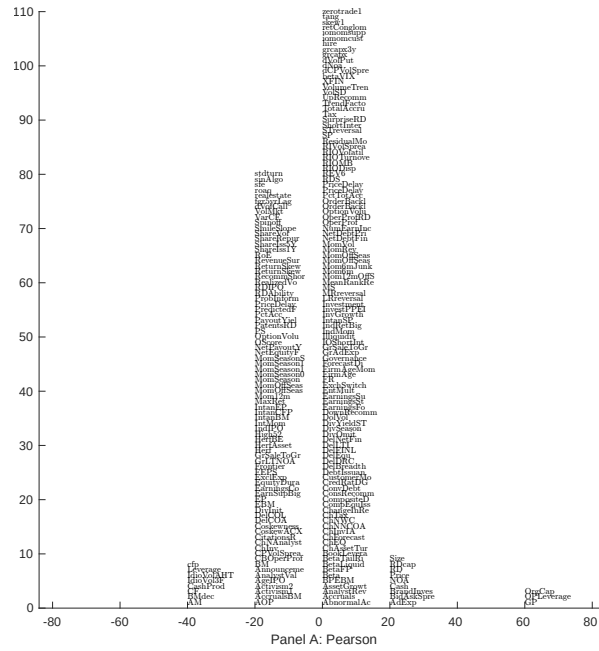


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OAE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

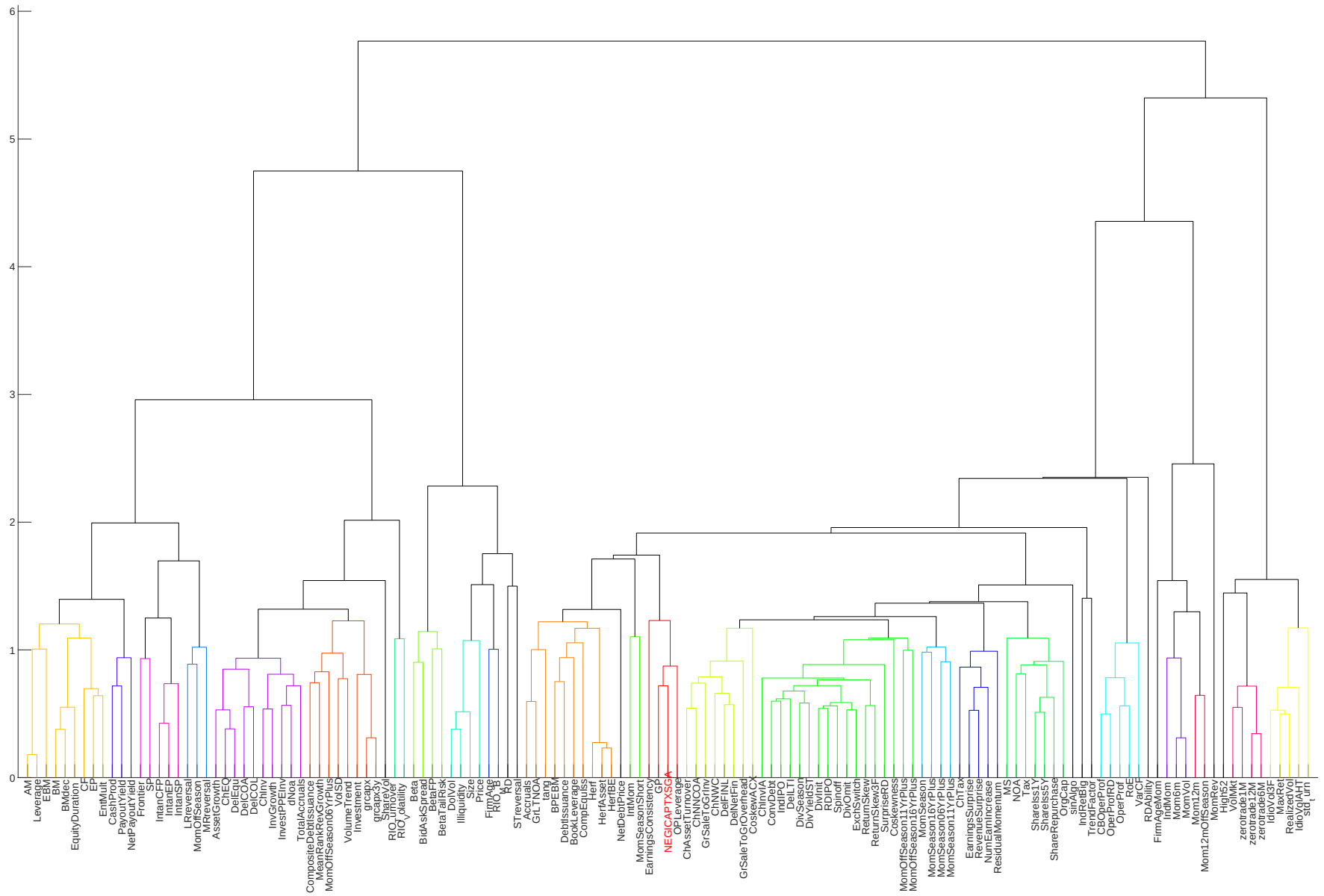


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

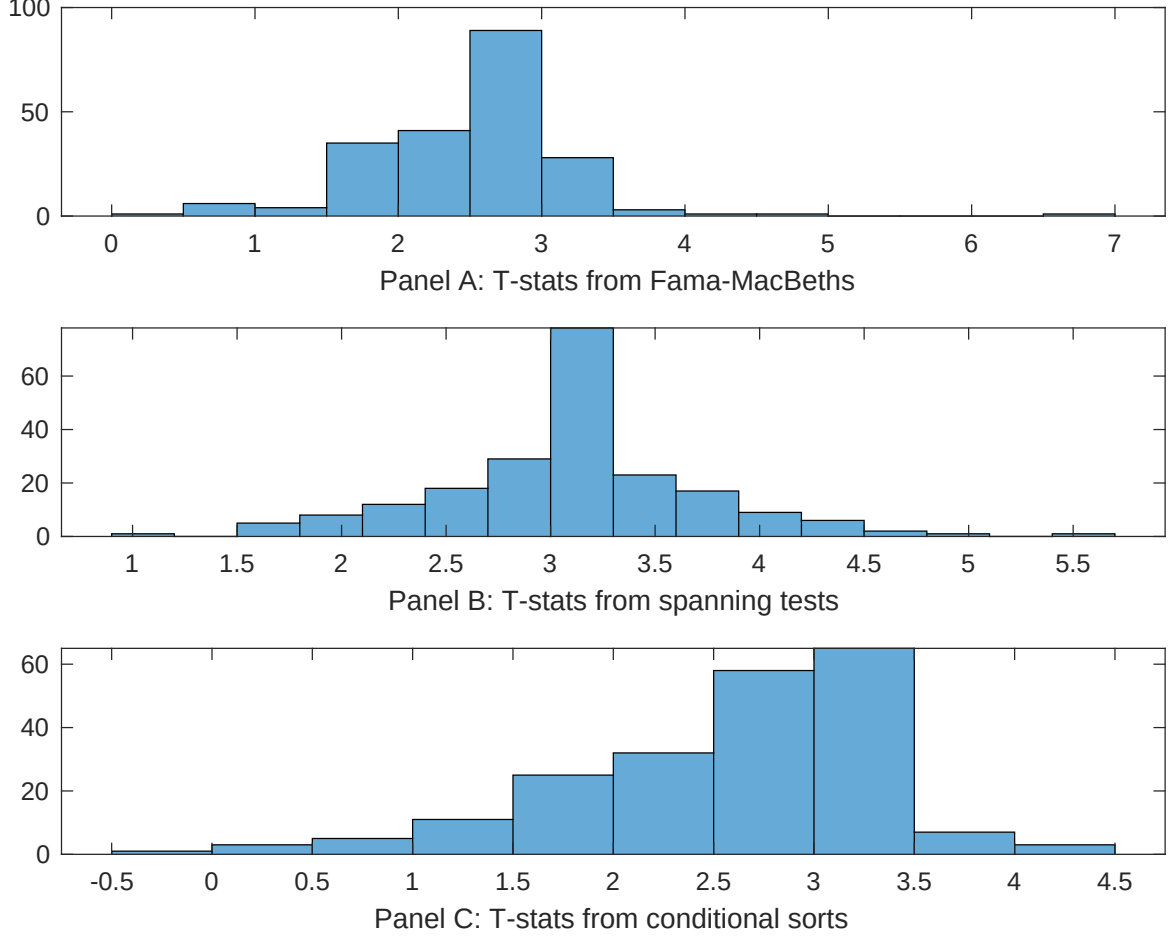


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OAE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OAE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OAE}OAE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OAE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OAE. Stocks are finally grouped into five OAE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OAE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OAE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OAE}OAE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.81 [2.85]	0.11 [4.70]	0.11 [4.82]	0.97 [4.12]	0.12 [5.21]	0.12 [5.01]	0.48 [1.77]
OAE	0.12 [0.21]	0.74 [1.69]	0.14 [2.94]	0.16 [3.43]	0.13 [2.68]	0.13 [2.74]	0.45 [0.88]
Anomaly 1	0.94 [5.93]						0.11 [7.45]
Anomaly 2		0.11 [2.96]					-0.48 [-1.34]
Anomaly 3			0.48 [3.26]				0.21 [2.42]
Anomaly 4				0.36 [5.64]			0.18 [1.58]
Anomaly 5					0.20 [1.97]		-0.45 [-0.58]
Anomaly 6						0.50 [2.88]	-0.19 [-1.99]
# months	732	732	727	727	727	727	727
$\bar{R}^2(\%)$	1	0	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OAE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OAE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.26 [2.72]	0.47 [4.19]	0.54 [4.54]	0.47 [4.15]	0.54 [4.55]	0.53 [4.48]	0.22 [2.41]
Anomaly 1	76.99 [19.54]						79.59 [15.94]
Anomaly 2		41.71 [9.26]					20.40 [4.55]
Anomaly 3			-2.16 [-0.43]				-13.24 [-0.90]
Anomaly 4				-54.43 [-7.48]			-25.49 [-3.99]
Anomaly 5					-13.77 [-2.35]		37.69 [6.93]
Anomaly 6						-6.48 [-1.33]	22.97 [1.60]
mkt	-13.08 [-5.71]	-11.48 [-4.29]	-11.74 [-4.03]	-11.25 [-4.12]	-13.42 [-4.66]	-11.14 [-3.83]	-10.15 [-4.29]
smb	2.58 [0.78]	-11.20 [-2.56]	8.10 [1.95]	26.07 [5.62]	8.66 [2.11]	8.57 [2.08]	-1.47 [-0.35]
hml	-24.00 [-5.06]	-35.03 [-6.15]	-53.88 [-6.74]	-0.49 [-0.05]	-46.21 [-6.64]	-49.05 [-6.33]	-24.31 [-2.88]
rmw	-17.94 [-3.58]	3.45 [0.60]	26.55 [4.78]	14.33 [2.57]	25.55 [4.63]	26.53 [4.81]	-33.67 [-6.27]
cma	10.05 [1.54]	-10.28 [-1.34]	-3.23 [-0.40]	0.41 [0.05]	-1.24 [-0.15]	-3.82 [-0.47]	4.79 [0.75]
umd	0.36 [0.16]	1.69 [0.63]	1.87 [0.60]	4.12 [1.52]	-0.59 [-0.19]	0.78 [0.25]	11.36 [4.31]
# months	732	732	728	728	728	728	728
$\bar{R}^2(\%)$	51	33	25	31	26	25	56

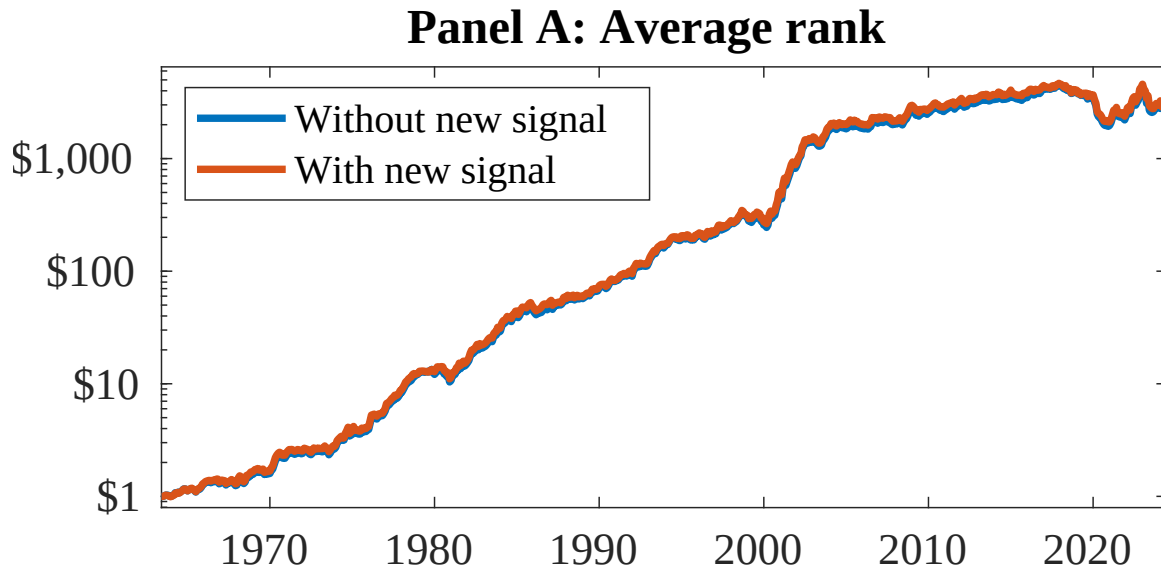


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OAE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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